# ARES: Autonomous Red-Planet Experimental Shelter

“Mars is there, waiting to be reached.” - Buzz Aldrin

**Theme:** Advanced Science Missions and Technology Demonstrators for Human-Mars Precursor Campaign
**University:** Columbia University


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# Columbia University
# *ARES: Autonomous Red-Planet Exploratory Shelter*

**Theme: Advanced Science Missions and Technology Demonstrators for Human-Mars Precursor Campaign**

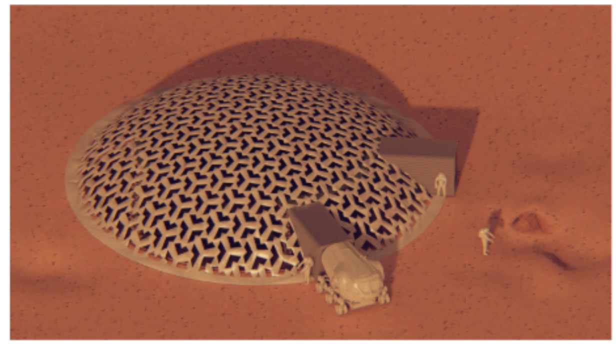
## Mission Objectives:

- Establish an initial human-based architecture prior to 2040
- Lower risks of high TRL technologies by proving their functionality in-environment
- Collect scientific data on regolith composition and availability of water-ice volatiles

## Technical Approach:

- Prospecting orbiter to assess landing site and lower risk
- Pressurized bistable auxetic dome to enable human presence
- Technical demonstration of Aqua Factorem and 3D printing systems

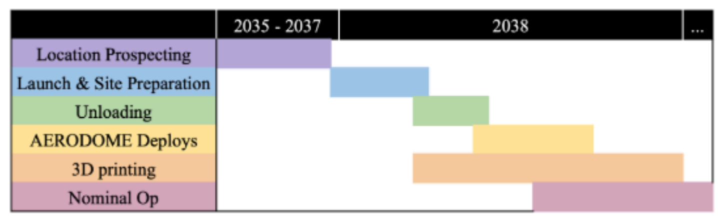
## Image:



## Key Design Details & Innovations:

- A bistable auxetic deployable dome for lightweight, compact launch and deployment
- Pressurized dome will provide a controlled environment for regolith-based 3D printing
- An adapted Aqua Factorem system will use excavator to extract water particles from regolith via vaporization
- The application of Kilopower nuclear reactors for power and thermal systems

## Timeline




## Budget:

| | |
|---|---|
| **Peak Power Generation: 80 kW** | **Mass: 76.314 T** |
| **Peak Power Draw: 78 kW** | **Volume: 542 m^3** |
| **Cost: $2,332,637,630.13** | |

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## I. Introduction

With the Artemis era of space exploration underway, the Moon is set to become a crucial gateway to the rest of the solar system, with Mars as the next frontier. Establishing a sustained human presence on Mars will require precursor missions to develop key infrastructure and ensure the survival of early crews. Among the many challenges of Martian settlement, the habitat itself is one of the most vital pieces of infrastructure, requiring extensive testing to ensure it can support human life in such an extreme environment. A successful Martian habitat must provide robust radiation shielding, reliable life support systems, efficient thermal management, and utilize local resources rather than relying on costly resupply missions from Earth.

Mission ARES is a technology demonstrator designed to validate novel habitat construction methods and essential support systems. The initial phase, launching in 2035, will feature an orbiter from the International Mars Ice Mapper mission, which will identify optimal landing sites to reduce mission risk. In 2037, a second launch will deliver the core mission payload, including a deployable, pressurized, triple-layer dome—demonstrating a modular, controlled habitat system. Inside that dome, a 3D-printed regolith-based concrete structure will provide work and living space, showcasing in-situ resource utilization (ISRU). This mission will collect data from the dome's interior to determine the feasibility of hydroponic farming within the structure. Key systems, such as NASA's MOXIE oxygen generator and an adapted version of the University of South Florida's Aqua Factorem water extraction system, will support long-term habitability.

This demonstration will validate a scalable, lightweight, and easily transportable habitat concept, proving Mars' readiness to host a human crew by the 2040s. If successful, a crew of four could move in with minimal additional systems, marking a critical step toward permanent human settlement on the Red Planet.

## II. Mission Overview and Timeline

The first phase of this mission is projected to launch in 2035. Since there is limited information about surface conditions on Mars, the goal of this initial phase is to determine an optimal location for a habitat fit for human survival and create a detailed map of topological characteristics. The unpredictability of Martian regolith currently poses uncertainties to the regolith excavator and 3D printing processes necessary for this mission, which require highly predictable terrain to function properly. This phase will also serve to gain scientific data on Martian terrain of scientific importance, particularly in the ice-rich regions of Arcadia Planitia and Erebus Montes.

The main phase will launch in 2037 aboard SpaceX's Starship, NASA's preferred superheavy rocket, which can easily carry our mission payload weighing about 76 metric tons. The lander will touch down at a site chosen based on data from the prospecting phase. Upon landing, four Surface Payload Early Assembly Rovers (SPEARs) will deploy via the unloading dock. Drawing inspiration from NASA's ATHLETE rover for load-bearing capabilities and Perseverance's arm dexterity, SPEARs will unload and set up mission components for assembly. Afterwards, they will conduct initial site analysis, compact and level the regolith, collect regolith samples, and refine I-MIM's topographic map. They will then unload and activate Kilopower nuclear reactors and install a system to distribute their waste heat.

Meanwhile, an Aqua Factorem rover will survey the area for ice-containing regolith, extract and process the water, and store it in designated containers. Then, a RASSOR excavator will exit via Starship's loading dock and the SPEAR crew will carry geopolymer concrete manufacturing equipment near the habitat site. After the SPEARs clear the site of debris and level the site area, they will lay an airtight floor for the Auxetic Exterior Radiation Obstructing Deployable Outpost for Martian Exploration (AERODOME) and position a compact robotic-arm 3D printer over it. Next, the AERODOME will be placed over the floor and printer, secured with drilled rods, and sealed for airtightness, requiring two weeks to cure. Then, compressed nitrogen gas will be released into the dome, expanding it into its deployed shape. MOXIE will produce oxygen, bringing the inner atmosphere to Earth-like conditions and composition over 70 days while dome heaters activate. Outside the dome, RASSOR will spend 4 months collecting regolith to create concrete. Combined with binders and water from Aqua Factorem, the concrete will be pumped into the dome to the 3D printer. Over the next 2 weeks, the Concrete Additively-Built Interior Nexus (CABIN) will be constructed. Further research on the efficiency of robotic activities and risk mitigation may result in buffers added to this timeline. As the mission operations are comfortably within the power budget, this would not dramatically change the mission [see Appendix F].

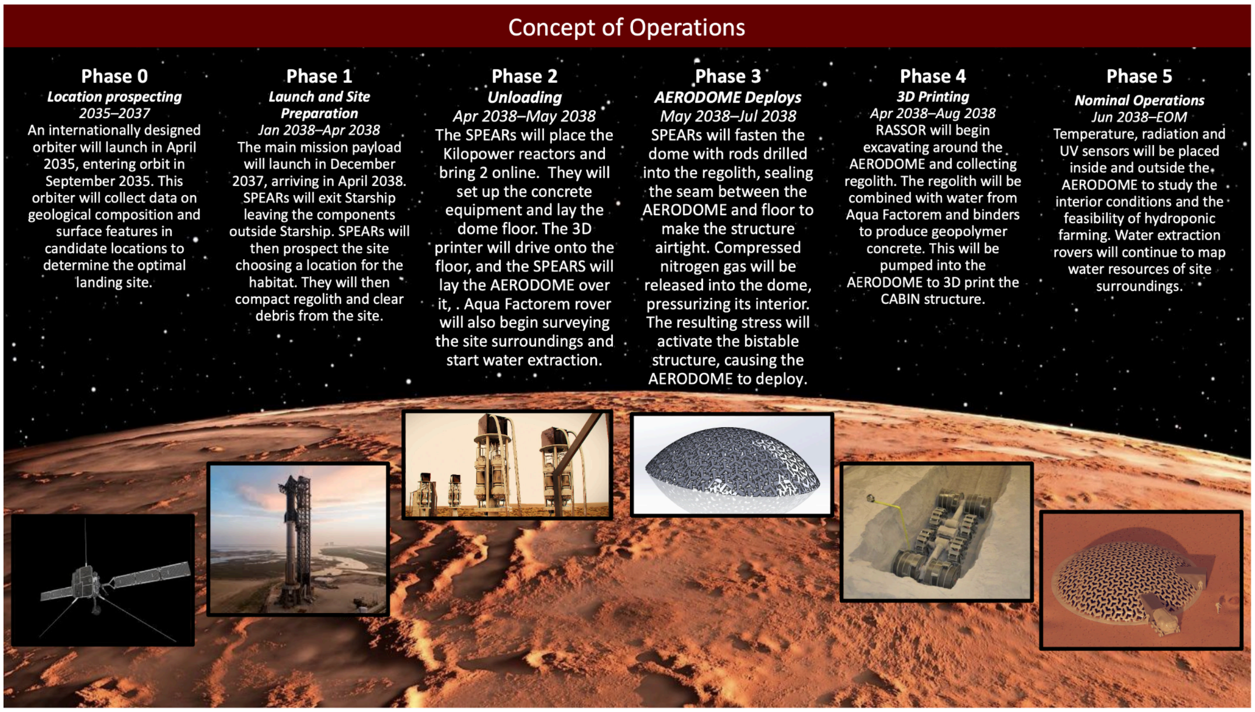


**Figure 1: CONOPS**

## III. Subsystems and Budgets

### A. Design Assumptions

The 2037 main launch informs several design assumptions. By then, we anticipate the Artemis missions will remain on schedule, and a lunar base capable of planetary launches will be operational. Additionally, we anticipate that significant investments will be made into the technologies necessary for this mission's success. Specifically for this mission design, our 3D printing, power, robotic autonomy technology, and water extraction systems will be advanced enough to be fully operational on the Moon and at the projected capabilities NASA has determined for aerospace contractors.

This mission will autonomously conduct operations while regularly transmitting data to Earth. NASA scientists can override autonomous programs when needed, though not for time-sensitive emergencies due to communication delays. Between phases, operations will pause for data review and verification from Earth before proceeding.

### B. Location Prospecting

The first phase of our mission, scheduled for 2035, will launch the International Mars Ice Mapper Mission (I-MIM) orbiter designed by NASA, the Agenzia Spaziale Italiana (ASI), the Canadian Space Agency (CSA), the Japan Aerospace Exploration Agency (JAXA), and the Netherlands Space Office

(NSO). The I-MIM is a low-orbit Mars orbiter carrying an L-band (930 MHz) polarimetric synthetic aperture radar (PolSAR) and nadir sounder to gather data on possible landing sites. Over the course of two Earth years, this orbiter will gather enough data on these four sites to safely choose a location for this mission and construct a detailed topographic map.

Research by SpaceX and JPL has identified four candidate sites for Starship: AP-1 and AP-9 (Arcadia Planitia), PM-1 (Phlegra Montes), and EM-16 (Erebus Montes), selected for their flat terrain, high thermal inertia, and accessible ice deposits. Existing radar data lacks resolution for the top 15–30 meters of regolith, leaving uncertainties about surface conditions. At an operating frequency of 930 MHz, the I-MIM's PolSAR 7-element feed array will penetrate up to 10-15 meters of dust and regolith, providing key information about Mars' surface roughness, regolith density, and cryosphere. This information would feed into a mapping algorithm to detail key topographic data and the location of ice deposits. In addition to ensuring a viable habitat location with ice deposits, I-MIM will address other high-priority scientific objectives on Mars: surface-atmosphere cycling, climate patterns, volcanic activity, shallow lava tubes, impact history, landforms, and geology.

International collaboration will distribute I-MIM's $185 million cost: CSA will provide PolSAR, JAXA the spacecraft bus, NSO and ASI the solar arrays, while NASA funds launch and communications for $5 million via Starship. Following the conclusion of the two Earth-year-long phases of the mission, final site selection and launch of the secondary mission phase will occur, preparing for a 2037 launch.

## C. Unpacking Plan

### C.1. SPEARs

The Surface Payload Early Assembly Rovers (SPEARs) are responsible for executing the primary tasks of unloading and positioning payloads for assembly. Powered by a rechargeable lithium-ion battery and a fuel cell battery, the SPEARs crew consists of four dynamic and dexterous rovers. Inspired by NASA's ATHLETE rover and Mars Perseverance, SPEARs will carry and manipulate heavy loads using an integrated tether system and robotic legs. They will also feature a Perseverance-like arm to operate tools like claws, drills, and sealant spreaders. The SPEAR's six legs can extend and angle to maintain their chassis stable and carry large loads. The ATHLETE rover's full-sized wheelbase has a 14-ton lifting capacity. For packing efficiency, we've halved the ATHLETE design, reducing lifting capacity to seven tons while scaling down power requirements. Although the expected lifting capacity is now 7 tons, we expect significant technology advancements to be made by the 2037 timeframe and assume that SPEARs will be able to carry at least 9 tons of cargo. This scaling also allows four SPEARs to fit in the rocket, minimizing the risk of failure from a single unit.

Once assembly is complete, the SPEARs will return to Starship for protection from dust storms and other hazards, entering hibernation mode. Non-essential functions will power down, and fuel cells will trickle-charge the lithium-ion batteries to prevent degradation. These strategies will enable the SPEARs to power up upon receiving a trigger signal and be used in future Martian activities.

### C.2. Unloading Plan

Our mission will utilize Starship's extended payload cargo space, which is 22m long including the rocket's cone nose. Using a Starship CAD model for dimensional accuracy and simple geometric shapes for each component's estimated volume, our payload will fit into Starship. As per Figure 2 demonstrates, the four SPEARs will be loaded upright on their wheels ready to unload cargo as soon as they land, with two of the crew being stacked atop each other for compact

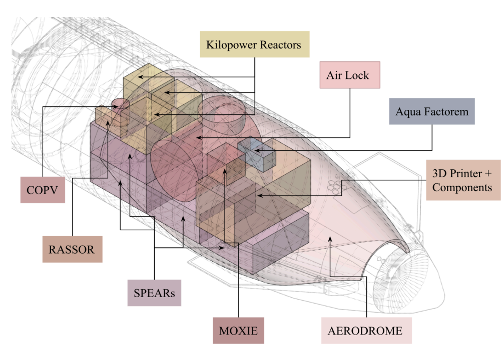


**Figure 2: Packing Plan**

packing. The AERODOME will be secured onto the Starship's inner walls using clips to prevent deformation during flight.

First, the rocket will land on its side, and the nose cone will then eject sideways releasing a ramp for the SPEARs to exit. The first SPEAR will exit carrying the 3D printing system, the Aqua Factorem rover, and the MOXIE system. The second SPEAR will follow carrying the dome's airlock. Next, the top SPEAR of the rover stack will step off the bottom SPEAR, carrying off the Kilopower reactors. The rovers will unload all their components with the help of the last rover that is not carrying anything near the starship landing. Except the SPEAR carrying the 3D printer, for safety that will be unloaded back into Starship. The SPEARs can then begin prospecting.

After prospecting, and choosing a location, the four SPEARs will now bring all components to their site starting with the dome. The clips securing the dome to the Starship's walls will be unactuated, and the four SPEARs will extend to their full 6m height to attach their tethers using their robotic arm, making a square base with each corner supported by a SPEAR. Then, the dome will slowly be pulled out by the SPEARS as they exit Starship until the dome has space to fully flatten on top of the SPEARs. At this point, the SPEARs will align the dome onto their four-corner base and carry it toward the site. After placing the dome, they will return to Starship to load the rest of the cargo onto themselves using their robust limbs, tether system, and each other and finish bringing everything to the site.

## D. AERODOME


Figure 3: Physical Prototype

The outer shell of the habitat system will consist of a bistable auxetic dome called the Auxetic Exterior Radiation Obstructing Deployable Outpost for Martian Exploration (AERODOME). Auxetic shapes are defined by their ability to stretch when placed under stress. Our design features a hexagonal sheet with a repeating geometric pattern containing triangular cutouts, which increase in size as they approach the dome's center. This configuration allows the dome to expand more centrally than at the edges, thereby causing it to lift upward. A current prototype shows proof of concept utilizing a laser cut rubber stamp sheet [Figure 3].

The dome's structural layer will be made from a continuous sheet of 10 cm thick thermoplastic polyurethane (TPU) with a laser-cut auxetic pattern. TPU's flexibility and strength make it ideal for Martian conditions, as it remains flexible at low temperatures while withstanding forces during dust storms. Its tensile strength, typically greater than 30 MPa, is sufficient to handle Earth-like internal pressures. However, to meet deployment and performance requirements, the TPU will need to be modified. The glass transition temperature must be well below -50°C to prevent embrittlement in Martian cold, and the brittle-to-ductile transition must occur at even lower temperatures. With TPU's versatility and ongoing research, we anticipate developing a specialized resin that meets these specifications, reaching a TRL of at least 7 by launch.

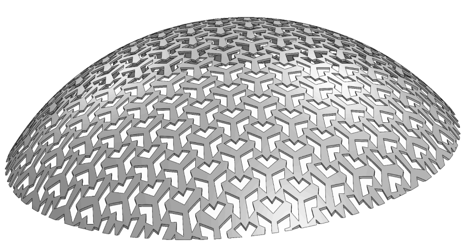
Figure 4: Auxetic Dome CAD

The protective layer above the TPU structural sheet will consist of optically transparent polyimide aerogels. Aerogels are known for their low density (0.15 g/cm³) and excellent thermal insulation properties (0.014 W/mK) and have been used on the Opportunity Rover. This polyimide aerogel combines the flexibility of polyimides and the optical transparency of silica aerogels. Silica aerogel testing shows that a 3 cm layer can reduce Martian UV-C flux from 3 W/m² to 50 nW/m² due to exponential transmittance decrease. Polyimide aerogel, with superior UV-C absorption from $\pi \rightarrow \pi^*$ transitions in chromophores, requires only 2 cm to meet insulation needs and safely lower UV radiation levels inside the dome. This layer will help maintain interior temperature, transmit sufficient visible light for hydroponics, and ensure human survival, with UV exposure inside the dome limited to 0.00432 J/m², much lower than NASA and ACGIH's 30 J/m² threshold. To protect the aerogel from

Martian dust storms and extend its durability, the dome will be covered with an ETFE (ethylene tetrafluoroethylene) layer, which offers excellent abrasion resistance and further reduces UV exposure with a 70% transmission rate.

The innermost layer of the dome is a clear TPU film about 3mm thick. This layer and the dome floor will create an airtight environment that will be used to pressurize and deploy the dome and maintain an internal pressure at about 1 atm. Once fastened, the SPEARs will circle the AERODOME's circumference, applying a 5cm wide band of two-component polyurethane sealant to create this airtight environment. The resin and hardener will be stored separately and mixed in small amounts over time as the application goes on. This type of sealant cures chemically, allowing it to cure in the Martian environment without additional work. The TPU film will withstand the required pressure, while the floor, made of 8 cm thick crosslinked extruded polyethylene (XPE) with 20% butylated pentaerythritol phosphate (BPP), provides the flexibility required for packing and the compressive strength to support the weight of the CABIN.

To initiate deployment, the flat, undeployed dome with all its layers will be laid over the 3D printer. Then the perimeter of the auxetic sheet will be fastened to the ground using a series of 0.5 m long steel rods bored into the regolith around the perimeter of the dome, 2 m apart. Then, nitrogen gas will slowly be released, increasing the interior pressure to about 0.78 atm. This pressurization will induce sufficient stress in the structural layer of the dome to activate the bistable structure, pushing the AERODOME upward from a flat sheet into a dome shape. This process is time- and energy-efficient compared to using stiffer materials. Additionally, the design eliminates challenges like dust-clogged hinges and provides better structural support, allowing for thicker radiation-blocking layers and greater durability in extreme Martian conditions.

To collect data about the effectiveness of the dome to create an environment suitable for plant growth and human survival, various sensors will be placed inside and outside the dome. Radiation and temperature sensors will compare levels to Mars's ambient conditions to determine the effectiveness of aerogel in this setting. Light sensors will compare the light levels inside and outside the dome to determine if the dome layers as well as weather patterns allow for plant growth.

To create openings for airlocks and resource locks, we will use zippered pressure seals bonded to the edges of the TPU sheet. In manufacturing, the hatch areas will be removed from all layers, and the zipper will be bonded using heat sealing to a new TPU sheet that fills the gap in the airtight AERODOME layer, ensuring a secure seal with the surrounding TPU. The dome will support two modified Lightweight External Inflatable Airlock (LEIA), which enable rover docking via the NASA Docking System (NDS). Although only one LEIA airlock will be launched with this mission, the dome will be able to support two airlocks for ingress/egress for future human habitation. Dual inflatable crew locks, reserved for emergency exits, will prevent Martian dust intrusion. SPEAR robotic systems will align the airlock with the zippered seal, after which it will pressurize its connection, forming an airtight bond. Internal vacuum pumps will remove residual air before equalizing pressure, ensuring a secure attachment before opening the zippered seal for LEIA access. The zippered seal then be opened, allowing interior access to LEIA. A similar mechanism will integrate oxygen and concrete production lines using sealed umbilical ports with independent pressure locks, featuring flexible, thermal, and dust-resistant gaskets for environmental control. LEIA modifications include adapting a docking port for dome compatibility and adjusting pressurization for Martian gravity. Currently at TRL 3, LEIA's development has broad applicability and is expected to advance before the mission.

**E. CABIN**

The Concrete Additively-Built Interior Nexus (CABIN) is designed to build the dome's interior structures for astronaut housing using Martian regolith and 3D Construction Printing (3DCP) technology. CABIN employs a 3D printer with a concrete binder, using ISRU materials from Mars. 3DCP technology is advancing rapidly, projected to grow from $300 million to $13 billion by 2032, with prototypes for space applications already in development. We anticipate it will reach TRL 6 by the mission launch, capable of autonomous printing.

We selected a 3D printer based on scalability and transportability. The printer must be expandable to meet shelter needs, compact for transport, and agile for deployment. Our design is inspired by the MaxiPrinter, a current 3D concrete printer, for its balance of these qualities. Its scorpion-inspired design allows it to navigate various terrains, while its compact size (3 m long and 2 m tall) ensures efficient storage and transport. Once deployed, it can print up to 7m from a stationary point.

CABIN will use geopolymer cement made from Martian regolith collected by the RASSOR Excavator. This alkali-activated cement requires minimal water, as compared to traditional concrete, and has shown good compressive strength in vacuums and extreme temperatures. Geopolymer cement contains aluminosilicate, calcium, magnesium, and iron minerals, with a water consumption of only 1.39%. The cement requires 29% activator and 37% solid aggregates. The activator, a sodium silicate, can be sourced from Martian regolith and atmosphere using ISRU, though sodium is more scarce and will be brought from Earth. Our design for the CABIN will use approximately 65 $m^3$ of concrete, requiring around 2,550 kg of the activators.

The CABIN system operates in stages. Upon landing, the RASSOR excavator will immediately deploy and begin collecting regolith. Regolith, water from Aqua Factorem, and activator are mixed in an insulated silo to create the concrete. Then, under the deployed AERODOME, the 3D printer starts construction. The concrete is then pumped to the printer extruder, which prints the CABIN structure The compact and expandable printer can navigate doorways and reach required distances, completing the print in about 12 days if run continuously.

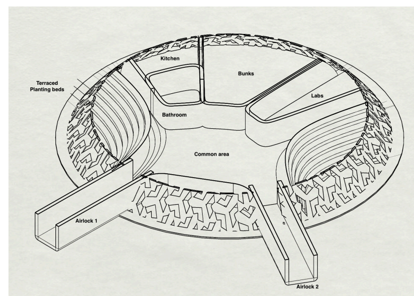


**Figure 5: Floor Plan**

**F. MOXIE**

NASA's MOXIE technology will produce $O_2$ to mix with $N_2$ to create a breathable atmosphere inside the AERODOME. The current small-scale MOXIE on Perseverance generates $O_2$ at 12 g/h; we plan to scale it up by 15 times with additional $CO_2$ collectors, $O_2$ storage, and thermal control systems. This enhanced MOXIE will achieve a fully breathable atmosphere within 70 days.

**G. Aqua Factorem**

Although Mars lacks any constant liquid water, several regions contain ice embedded in the regolith. Our landing site will likely offer abundant ice deposits and a traversable landscape for efficient extraction. If we assume that Aqua Factorem has been tested and used in space-simulated environments and the Moon, it is very likely that we can implement this system on Mars. Aqua Factorem is a concept for lunar water extraction that filters crystalized ice from the lunar regolith and vaporizes it, so that the result is purified water and the potential for hydrogen fuel. The water extraction process consists of excavation, rock and gravel separation, grinding and fracturing, beneficiation, pneumatic and magnetic separation, and finally tribocharging and electrostatic separation. Essentially, Aqua Factorem's extraction design consists of sorting and fracturing the regolith to collect soil that would be put through various electrically charged separation, vaporization, and purification processes to retrieve our water.

Aqua Factorem can be adapted for Mars with minor adjustments. A key modification is the addition of an infrared scanner to differentiate between water ice and dry carbon dioxide ice. To improve rover autonomy, components from previous Martian rovers, such as hazard cameras, computing elements, and lighting, will be integrated. Unnecessary systems, such as magnetic and electrostatic separation, will be removed since the goal is to extract only water. The beneficiation process will be adjusted to remove excess minerals instead of collecting them.

The Martian Aqua Factorem will be powered by lithium-ion batteries recharged by Kilopower reactors and require approximately 39 kW to operate. The rover will autonomously produce 77.5 kg of water daily, storing it in beneficiation storage vessels. Power consumption and water production will vary with Martian seasons; summer will reduce power costs because the ice regolith is at a higher temperature, increasing water collection, while the colder seasons will have the opposite effect. Over time, water will accumulate in storage tanks for future missions and technology demonstrations.

The primary goal is to produce water for geopolymer cement. Aqua Factorem will produce the necessary in one month, but the accumulated water can also be used in production during this period. Over time after the 3D printing is finished, our water collection will begin to stockpile within its storage tanks for future crewed missions.

**H. Power Generation**

NASA is advancing nuclear fission surface power to provide uninterrupted energy. In 2018, NASA and the Department of Energy's National Nuclear Security Administration (NNSA) successfully demonstrated the Kilopower Reactors Using Stirling Technology (KRUSTY) system at TRL 6, achieving a stable 1 kW output over 28 hours. With continued investment, NASA expects to develop a 40 kW output under 6 metric tons by 2030. For the mission's peak draw of 78 kW, two KRUSTY reactors, each producing 40 kW (totaling 80 kW), will be sufficient, with one additional reactor for redundancy.

While alternatives like solar, high-atmosphere wind turbines, geothermal, and ground wind energy were explored, they proved unsuitable for Mars' environment due to challenges such as months-long dust storms, thin atmosphere, and the planet's deep crust. These factors make renewable options unreliable, reinforcing the choice of nuclear fission, which provides a consistent and dependable energy output.

For our fuel needs, it's important to note that a KRUSTY core, with an optimal engine configuration of eight 250-We Stirling engines, can produce 8 kWt. KRUSTY was designed to maintain a near-constant temperature over large power loading and fluctuations. For that reason, we estimate a 40kW output from a similar Kilopower may be achievable with a similar fuel amount via design optimization between system thermal resistance and Stirling engine temperature. We estimate that 35 kg of fuel per reactor will be sufficient for this mission. Paired with a complete engine array, KRUSTY, in its full capabilities, is intended to produce about 8 kW of thermal energy at 800 °C. Therefore, the conversion rate is 8 kWt to 2 kWe. With this conversion rate, our 40 kWe Kilopower reactors would produce about 160 kW of thermal energy. This means a waste of 120 kWt per reactor, or 240 kWt in total.

To maximize efficiency, we will use the waste heat generated by the Kilopower reactors for habitat heating. Heat exchangers in the reactor's cooling system will redirect the thermal energy, which will travel through insulated piping to the habitat. This system will regain 6 kW of otherwise wasted thermal energy from 2 Kilopower reactors, reducing the power necessary to heat the habitat by about 25%.

**I. Communication Systems**

Mission ARES will use NASA's Space Communications and Navigations (SCaN) program, which supports the ISS, Mars rovers, and the Artemis mission. All communication and command handling from Earth will occur through the Central Communication Hub (CCH), equipped with a UHF antenna for primary communication with the Mars Relay Network (MRN), providing near-continuous coverage of Mars. Key systems like the SPEARs, MOXIE, Aqua Factorem, CABIN 3D printer, RASSOR, and AERODOME will have UHF antennas to communicate with the CCH. Communication within the habitat will also pass through the CCH, with Earth-based commands processed and sent to the respective systems, while data such as AERODOME environmental analyses will be transmitted back to Earth. In case of MRN coverage gaps, the CCH will also have an X-band antenna for direct communication with the Deep Space Network (DSN) on Earth. Redundant antennas on the CCH ensure continued communication in case of failure.

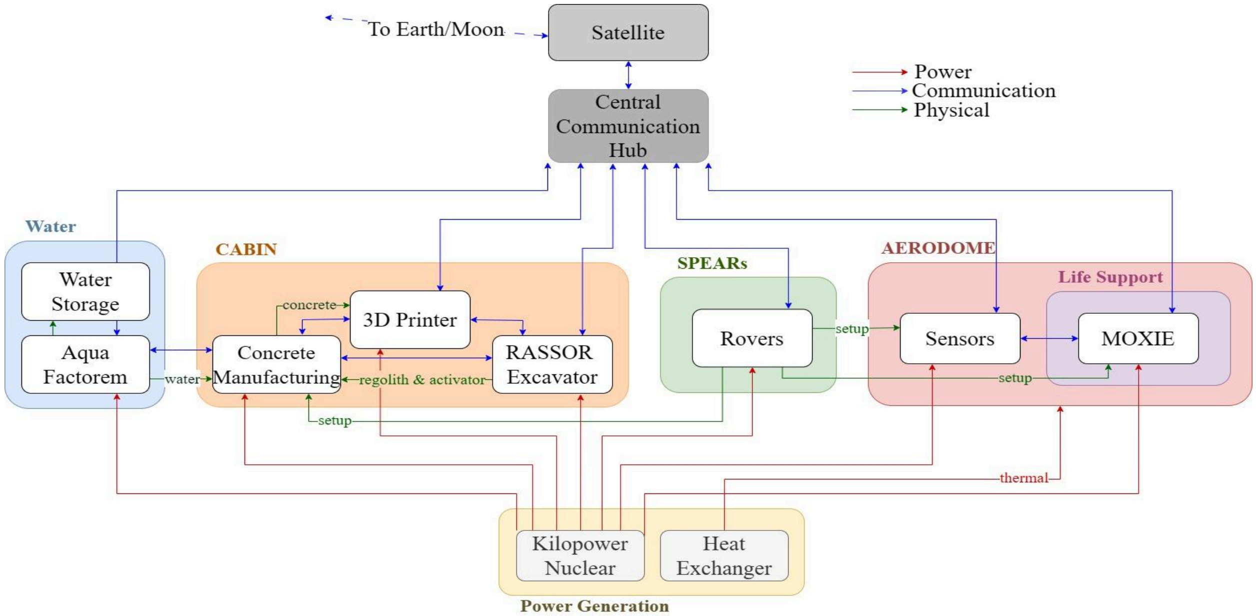


**Figure 6: Communications Block Diagram**

**J. Budget**

| Component | Total Power (kW) | Total Mass (kg) |
|---|---|---|
| **AERODOME** | -24.8 | 49305.89 |
| **Drilling and 3D Printing** | -20.791 | 4025 |
| **Water Extraction** | -38.8778 | 2433.952 |
| **SPEARs** | -9 | 1880 |
| **Power** | 83.5312 | 18570 |
| **Total Mass** | **76.314 Tons** | |
| **Peak Draw: 78 kW** | **Peak Generation: 80 kW** | |

**Table 1 : Budgets Summary**

Mission ARES will cost a total of $2.33 billion over the fiscal years 2025-2037. This budget accounts for hardware R&D as well as manufacturing from both NASA and industry contractors. In NASA's 2023 budget request for the 2025 fiscal year, the agency is requesting about $8 billion for Moon to Mars Systems, and is requesting increased funding for the years to follow. Assuming the Artemis missions follow their completion trajectory and that the Agency's funding remains the same, our budget analysis falls safely within NASA's budget request for this time frame. Mission ARES cost would be less than 3% of NASA's Moon to Mars budget over the next 12 years.

Our AERODOME is the most expensive portion of our mission budget simply due to its experimental nature; we implemented a 20% margin to this subsystem for further research and development. Other experimental components with more developed technologies, such as the Aqua Factorem and SPEARs, have a 10% margin in our budget analysis for further testing. Other system budget estimates are derived from current industry pricing for Starship launches, I-MIM deployment, 3D regolith printing, and Kilopower generators, all with a 5% margin. We anticipate cost fluctuations as technology readiness levels and system production advance.

| Component | Current TRL | Mitigation |
|---|---|---|
| Bistable Auxetic Structure | 3 | Scaling & testing within relevant environment |
| MOXIE | 5 | Testing within architecture |
| RASSOR | 4 | Development & testing |
| Regolith 3D Printer | 3 | Research & Development in relevant environment |
| LEIA | 4 | Integration & scaling |
| Aerogel | 5 | Testing within relevant environment |
| TPU | 4 | Modifications, Testing, and Integration |

**Table 2: Technology Ready Levels**

## IV. Long Term Viability and Risk Mitigation

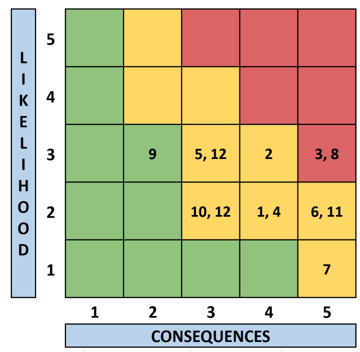


**Figure 7: Risk Matrix**

Our Mission minimizes risk by implementing high-TRL technologies. The first-phase orbiter will reduce mission uncertainty by gathering critical landing site data. The 3D printing and RASSOR excavating will allow for swift and efficient structural repairs to the dome when needed. Further research will ensure the AERODOME's structural stability against dust storms and Marsquakes, while external vehicles like SPEARs will take shelter inside Starship to prevent dust damage. A robust egress architecture ensures astronaut safety in emergencies, including fires, leaks, and power failures. Redundant communication antennae will maintain reliable contact within the habitat and with Earth via the Mars Relay Network. While maintaining the technology development schedule is crucial, the habitat's demonstrative nature mitigates risks if full TRL readiness isn't achieved on time. Major mission risks are listed in the table below along with a risk matrix to categorize them by level of criticality. Based on the risk matrix, the team will focus on developing the dome technology to ensure that it may reliably expand and protect from the Martian environment.

## V. Conclusion

Mission ARES demonstrates solutions to a vast number of technology shortfalls NASA currently faces for Mars crewed missions. The preliminary orbiter phase of the mission sets the stage for healthy international collaboration, spreading the total risk and costs of the mission across several partner countries. The use of bistable auxetic materials in the aerospace industry is unprecedented, and our implementation of these unique designs will generate various solutions in volume, autonomous deployments, and modular expansion on Mars and the Moon. All other mission subsystems mitigate risk through their operations and design, while also putting an emphasis on data collection for risk prevention in future missions. With proper execution and development, future mission steps could incorporate international partnerships and ensure a safe, peaceful Mars settlement.

While we have identified and demonstrated a promising concept for a deployable Martian shelter, we intend to develop a fully refined model to conduct Finite Element Analysis (FEA) and thermal analysis. For a physical prototype, the ARES team will explore other AERODOME materials and manufacturing processes (such as TPU 3D printing) for strength and compression tests. In order to ensure a successful mission, advancements in technologies and human lunar presence from now until the next decade must be made in a timely manner. If continued collaboration increases and technologies advance, Mission ARES will be the first precursor mission to a larger Martian campaign.

## VI. Appendix

### Appendix A: References


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**Appendix B: Money Budgets**

| | Quantity | Unit Cost | Margin | Total Cost |
|---|---|---|---|---|
| **Launch** | **2** | | | **$321,100,000.00** |
| Launch | 1 | $65,000,000.00 | 30.00% | $84,500,000.00 |
| Starship Launch | 1 | $182,000,000.00 | 30.00% | $236,600,000.00 |
| **AERODOME** | **53** | | | **$1,081,499,367.19** |
| Structural Auxetic Layer (TPU) | 1 | $25,000,000.00 | 20.00% | $30,000,000.00 |
| Fastening Rods | 40 | $10.27 | 5.00% | $431.34 |
| Electric Resistance Heater | 1 | $40,000.00 | 5.00% | $42,000.00 |
| ETFE layer | 1 | $30,177.00 | 5.00% | $31,685.85 |
| Aerogel layer | 1 | $3,300,000.00 | 5.00% | $3,465,000.00 |
| Airtight layer (clear TPU film) | 1 | $400,000.00 | 10.00% | $440,000.00 |
| Crosslinked polyethylene foam (XPE) with BPP floor | 1 | $2,000,000.00 | 10.00% | $2,200,000.00 |
| UV sealant | 1 | $5,000.00 | 5.00% | $5,250.00 |
| Hatches | 2 | $15,000.00 | 5.00% | $31,500.00 |
| Pressurized Nitrogen | 1 | $20,000.00 | 5.00% | $21,000.00 |
| Nitrogen Composite Tank (COPV) | 1 | $250,000.00 | 5.00% | $262,500.00 |
| AIRLOCKS | 1 | $200,000,000.00 | 10.00% | $220,000,000.00 |
| MOXIE | 1 | $750,000,000.00 | 10.00% | $825,000,000.00 |
| **Location** | **1** | | | **$5,250,000.00** |
| Communication Systems | 1 | $5,000,000.00 | 5.00% | $5,250,000.00 |
| **Drilling and 3D Printing** | **3** | | | **$41,365,435.00** |
| 3D Concrete Printer, Pump and Mixer | 1 | $34,000,000.00 | 20.00% | $40,800,000.00 |
| Alkali Activators (Sodium) | 1 | $14,700.00 | 5.00% | $15,435.00 |
| Excavator | 1 | $500,000.00 | 10.00% | $550,000.00 |
| **Water Extraction** | **24** | | | **$514,330,997.50** |
| Excavator | 4 | $500,000.00 | 10.00% | $2,200,000.00 |

| | | | | |
|---|---|---|---|---|
| Aqua Factoreum | 1 | $213,000,000.00 | 10.00% | $234,300,000.00 |
| Cachecam | 2 | $18,900,000.00 | 5.00% | $39,690,000.00 |
| Flood lights | 4 | $190.00 | 5.00% | $798.00 |
| Hazard Camera | 8 | $18,900,000.00 | 5.00% | $158,760,000.00 |
| Navigation Camera | 4 | $18,900,000.00 | 5.00% | $79,380,000.00 |
| Rechargeable Lithium Batteries | 1 | $190.00 | 5.00% | $199.50 |
| **SPEARs** | **8** | | | **$345,598,000.00** |
| Locomotion | 4 | $69,245,000.00 | 10.00% | $304,678,000.00 |
| Robotic Internals | 4 | $9,300,000.00 | 10.00% | $40,920,000.00 |
| **Power** | **9** | | | **$143,175,000.00** |
| Power Generation and Heat Routing | 3 | $20,000,000.00 | 10.00% | $66,000,000.00 |
| Total fuel | 6 | $12,250,000.00 | 5.00% | $77,175,000.00 |
| **Total** | | | | **$2,452,318,799.69** |

**Appendix C: Power and Mass Budgets**

| | | Power | | Mass | |
|---|---|---|---|---|---|
| Component | Quantity | Unit Power (kW) | Total Power (kW) | Unit Mass (kg) | Total Mass (kg) |
| **AERODOME** | | | **-24.8** | | **49305.89** |
| Structural Auxetic Layer (TPU) | 1 | 0 | 0 | 35400 | 35400 |
| Fastening Rods | 40 (2 at a time) | -0.9 | -1.8 | 1.046 | 41.84 |
| Electric Resistance Heater | 1 | -18.5 | -18.5 | 60 | 60 |
| ETFE Layer | 1 | 0 | 0 | 1940 | 1940 |
| Aerogel Layer | 1 | 0 | 0 | 1500 | 1500 |
| Airtight Layer (Clear TPU Film) | 1 | 0 | 0 | 880 | 880 |
| Crosslinked Polyethylene Foam (XPE) with BPP Floor | 1 | 0 | 0 | 1100 | 1100 |
| UV sealant | 1 | 0 | 0 | 65 | 65 |
| Hatches | 2 | 0 | 0 | 0.4 | 0.8 |
| Pressurized Nitrogen | 1 | 0 | 0 | 900 | 900 |
| Nitrogen Composite Tank (COPV) | 1 | 0 | 0 | 1000 | 1000 |
| AIRLOCKS | 1 | 0 | 0 | 5861.25 | 5861.25 |
| MOXIE | 1 | -4.5 | -4.5 | 557 | 557 |
| **Drilling and 3D Printing** | | | **-20.791** | | **4125** |
| 3D Concrete Printer | 1 | -7 | -7 | 2500 | 2500 |
| Pump and Silo | 1 | -12 | -12 | 750 | 750 |
| Alkali Activators (Sodium) | 1 | 0 | 0 | 850 | 850 |
| Excavator | 1 | -1.791 | -1.791 | 25 | 25 |
| **Water Extraction** | | | **-38.8778** | | **2433.952** |
| Excavator | 4 | -1.791 | -7.164 | 25 | 100 |
| Beneficiation System | 1 | -15 | -15 | 100 | 100 |

| | | | | | |
|---|---|---|---|---|---|
| Crusher | 1 | -4.932 | -4.932 | 50 | 50 |
| Transporter | 1 | -0.0736 | -0.0736 | 25 | 25 |
| Regolith Binding Agent | 1 | 0 | 0 | 10 | 10 |
| Power/Thermal/Control System | 1 | -0.9 | -0.9 | 1500 | 1500 |
| Melter | 1 | -3.829 | -3.829 | 3.5 | 3.5 |
| Degasification | 2 | -0.0736 | -0.1472 | 3.5 | 7 |
| Reverse Osmosis | 1 | -4.932 | -4.932 | 10.5 | 10.5 |
| Catalytic Carbon Filter | 1 | -0.0736 | -0.0736 | 3.5 | 3.5 |
| Storage Vessels | 1 | 0 | 0 | 550 | 550 |
| Cachecam | 2 | -0.1 | -0.2 | 0.425 | 0.85 |
| Flood Lights | 4 | -0.4 | -1.6 | 11 | 44 |
| Hazard Camera | 8 | -0.0022 | -0.0176 | 0.245 | 1.96 |
| Navigation Camera | 4 | -0.0022 | -0.0088 | 0.22 | 0.88 |
| Rechargeable Lithium Batteries | 1 | 12 | 12 | 26.762 | 26.762 |
| **SPEARs** | | | **-9** | | **1880** |
| ATHLETE Wheel Base | 4 | -2.2 | -8.8 | 425 | 1700 |
| PERSEVERENCE arm | 4 | -0.05 | -0.2 | 45 | 180 |
| Lithium-Ion Battery | 4 | 12 | 48 | 26.762 | 107.048 |
| Fuel Cells | 4 | 1.5 | 6 | Included in Athlete Base | Included in Athlete Base |
| **Power** | | | **89** | | **12570** |
| Kilopower Reactors | 2 | 40 | 80 | 6000 | 12000 |
| Heat Routing | 3 | 3 | 9 | 120 | 360 |
| Total Fuel | 6 | 0 | 0 | 35 | 210 |
| **Total** | | | **78 (Peak Draw)** | | **70314.842** |
| | | | **80 (Peak Generation)** | | |

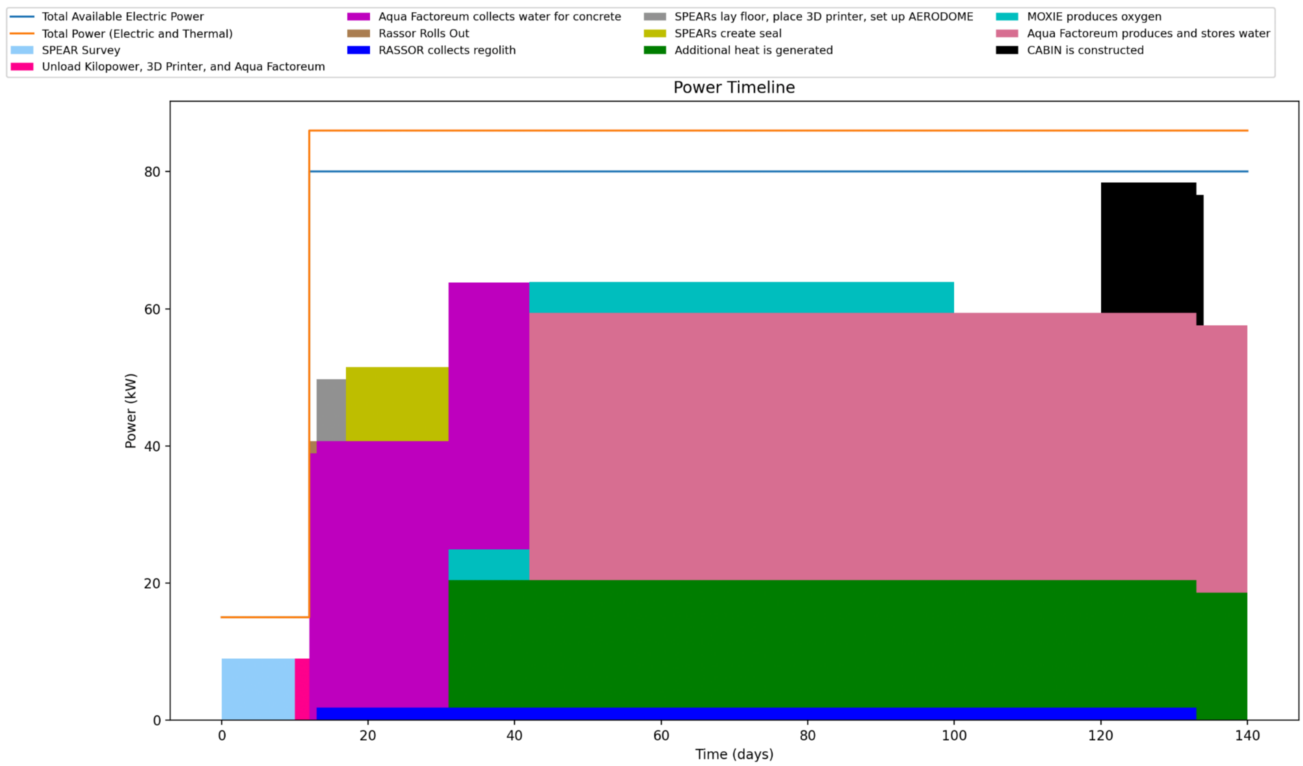
**Appendix D: System Technological Readiness Levels**

| System | Component | TRL | Justification/Mitigation |
|---|---|---|---|
| I-MIM | Orbiter | 7 | Fully operational in Earth applications |
| AERODOME | Bistable Auxetic Structures | 3 | Scaling & testing within relevant environment |
| AERODOME | Auxetic TPU | 4 | Modifications, Testing, and Integration |
| AERODOME | Aerogel | 5 | Testing within relevant environment |
| AERODOME | Nitrogen Composite Tank (COPV) | 8 | Validation through successful mission operations |
| AERODOME | ETFE (ethylene tetrafluoroethylene) | 6 | Scaling and testing within architecture |
| AERODOME | Extruded Polyethylene (XPE) | 4 | Development and testing within relevant environment |
| AERODOME | Butylated Pentaerythritol Phosphate (BPP) | 4 | Development and testing within relevant environment |
| AERODOME | MOXIE | 5 | Testing within architecture |
| AERODOME | LEIA | 4 | Integration & scaling |
| AERODOME | Zippered Pressure Seals | 7 | Testing within architecture |
| AERODOME | Two-component polyurethane sealant | 3 | Research & Development in relevant environment |
| CABIN | Geopolymer cement | 6 | Testing within architecture |
| CABIN | RASSOR | 4 | Development & testing |
| CABIN | 3D printing | 3 | Research & Development in relevant environment |
| Aqua Factorem | Aqua Factorem | 3 | Research & Development in relevant environment |
| SPEAR | SPEAR | 3 | Integration and Testing |
| Power | Lithium Ion Batteries | 8 | Validation through successful mission operations |
| Power | Fuel Cell | 8 | Validation through successful mission operations |
| Power | Kilopower Reactors | 4 | Scaling and testing |
| Communications | Mars Relay Network | 8 | Validation through successful mission operations |
| Communications | UHF Antennas | 8 | Validation through successful mission operations |

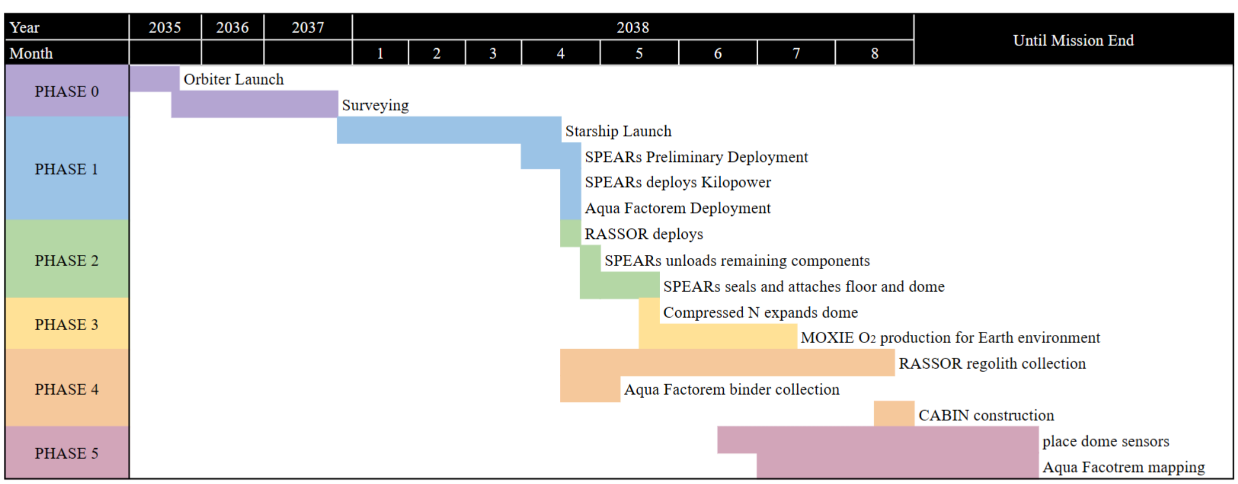
**Appendix E: Risk Mitigation Table**

| ID | Summary | L | C | Approach | Risk Statement | Status |
|---|---|---|---|---|---|---|
| 1 | Power | 2 | 4 | Accept | Power system fails to general necessary power | Active |
| 2 | Thermal | 3 | 4 | Mitigate | Sufficient heat is not generated within dome structure | Active |
| 3 | Mechanical | 3 | 5 | Research | Dome is unable to expand | Active |
| 4 | Communication | 2 | 4 | Mitigate | Failure of transceiver to communicate with orbiter | Active |
| 5 | Schedule | 3 | 3 | Watch | Required TRLs are not achieved by 2035 | Active |
| 6 | Mechanical | 2 | 5 | Mitigate | Marsquake causes dome failure | Active |
| 7 | Thermal | 1 | 5 | Research | Fire starts within the dome | Active |
| 8 | Mechanical | 3 | 5 | Mitigate | Leak in dome seal lets in dust and releases air and pressure | Active |
| 9 | Mechanical | 3 | 2 | Research | Dome does not sufficiently protect humans from radiation | Active |
| 10 | Mechanical | 2 | 3 | Mitigate | 3D printing mechanism malfunctions | Active |
| 11 | Mechanical | 2 | 5 | Research | Dome structure mechanical failure | Active |
| 12 | Communication | 3 | 3 | Mitigate | Failure to communicate within habitat | Active |
| 13 | Power | 2 | 5 | Mitigate | Nuclear energy leaks radioactive waste into environment | Active |

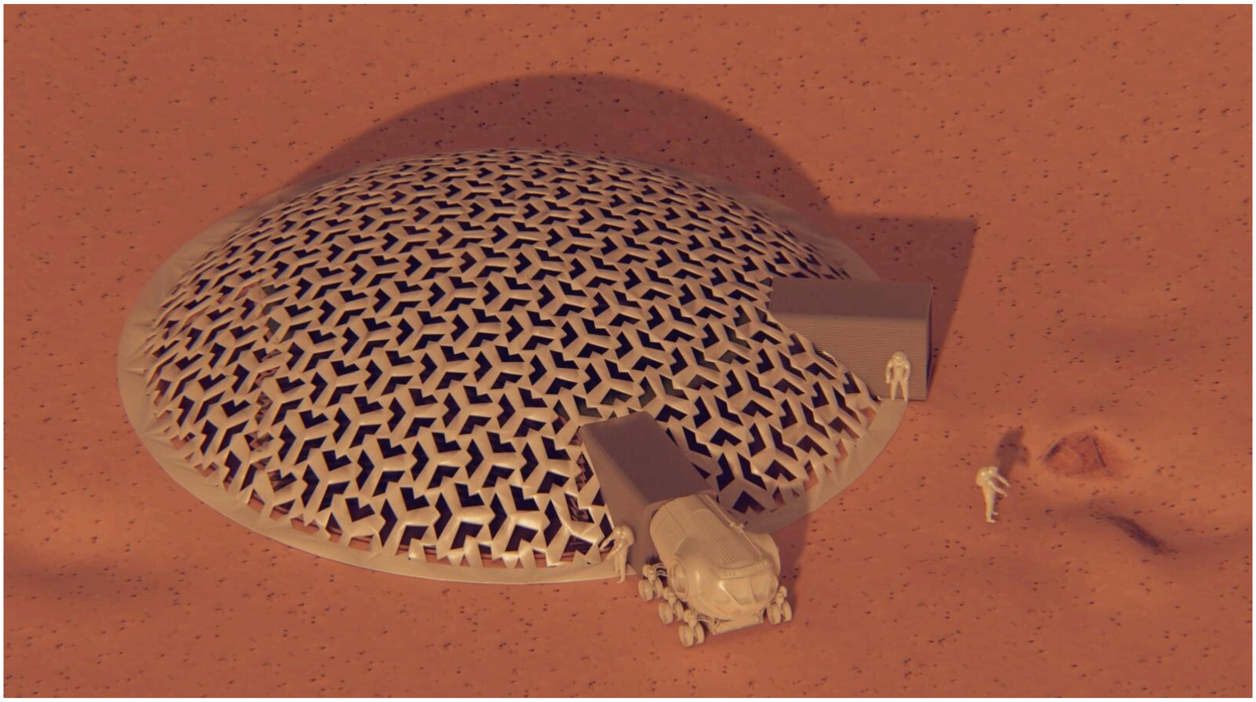
## Appendix F: Power Timeline

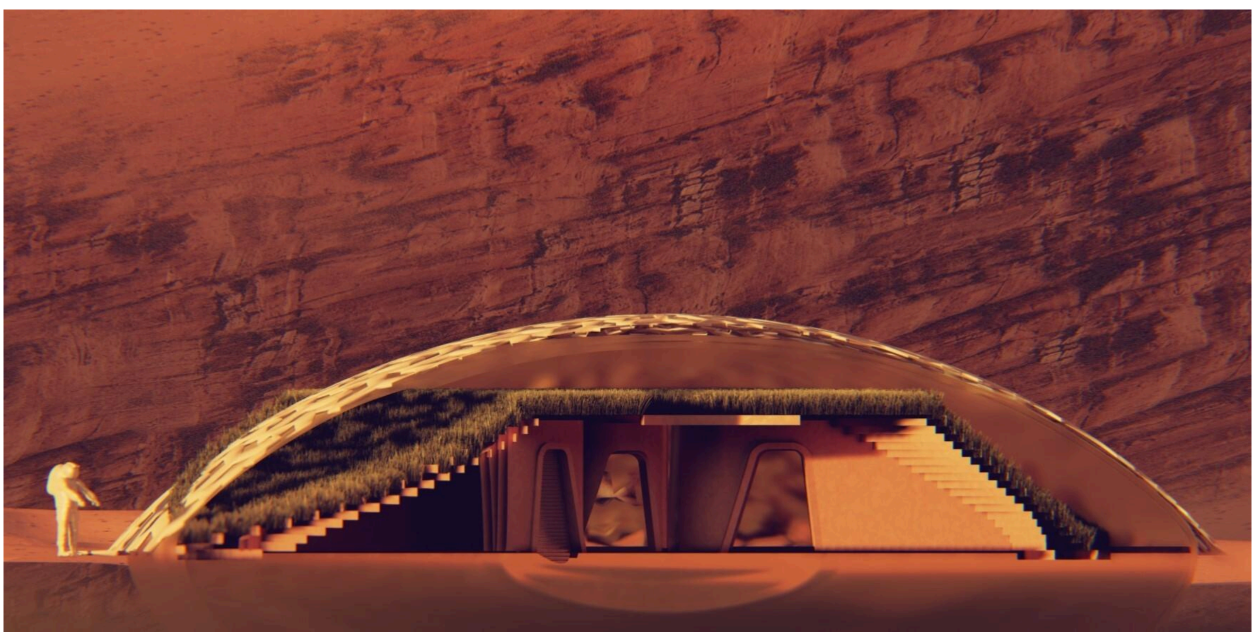


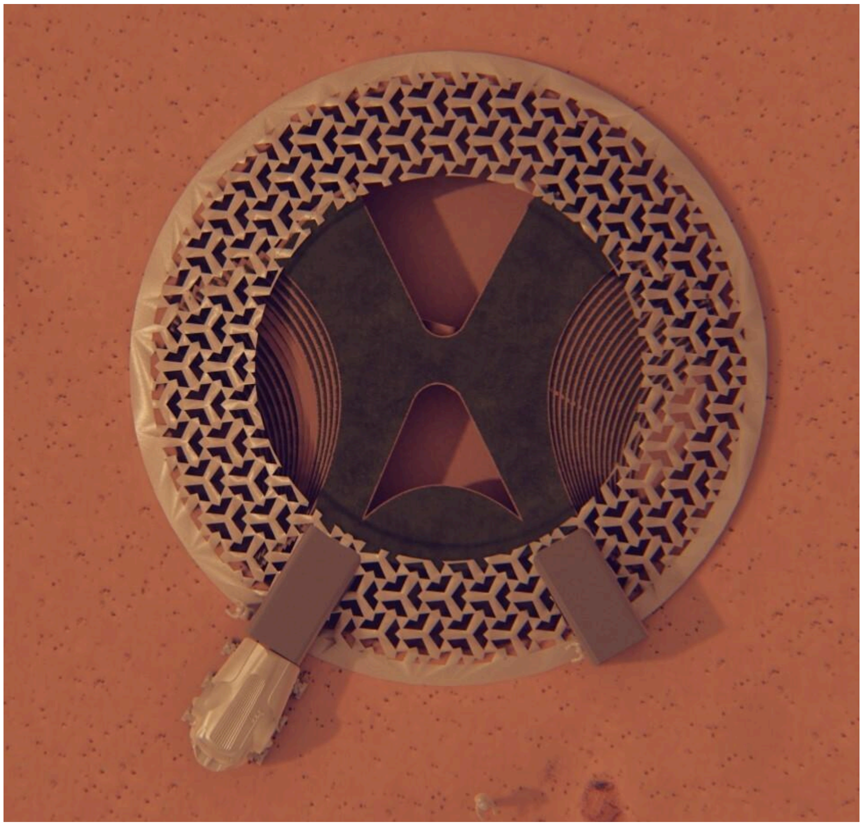

## Appendix G: Mission Timeline

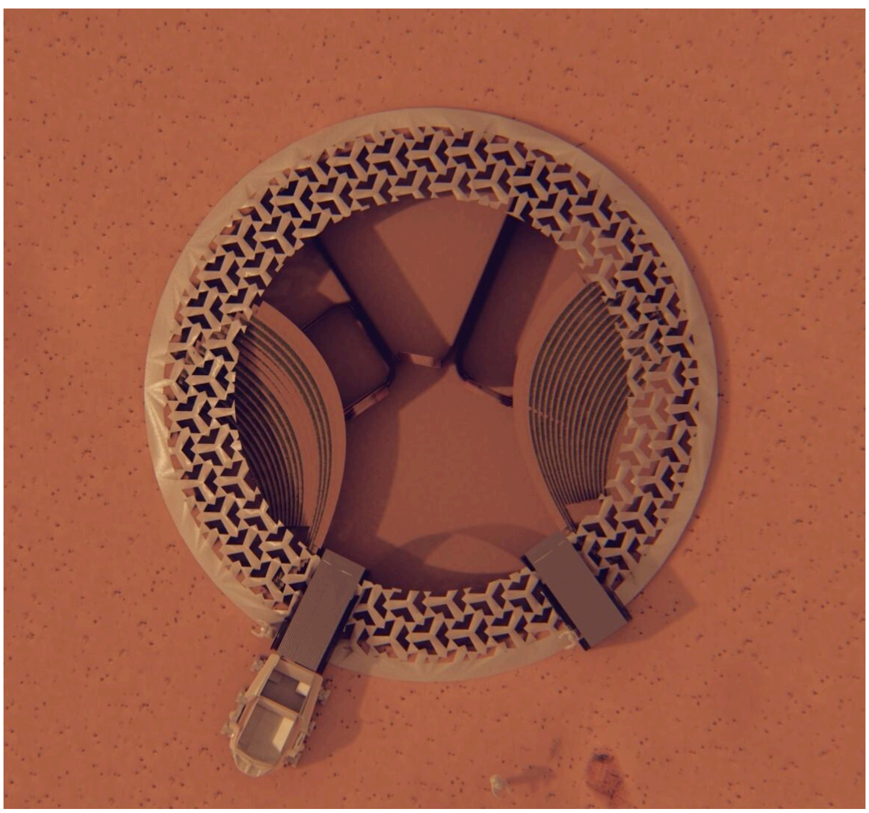


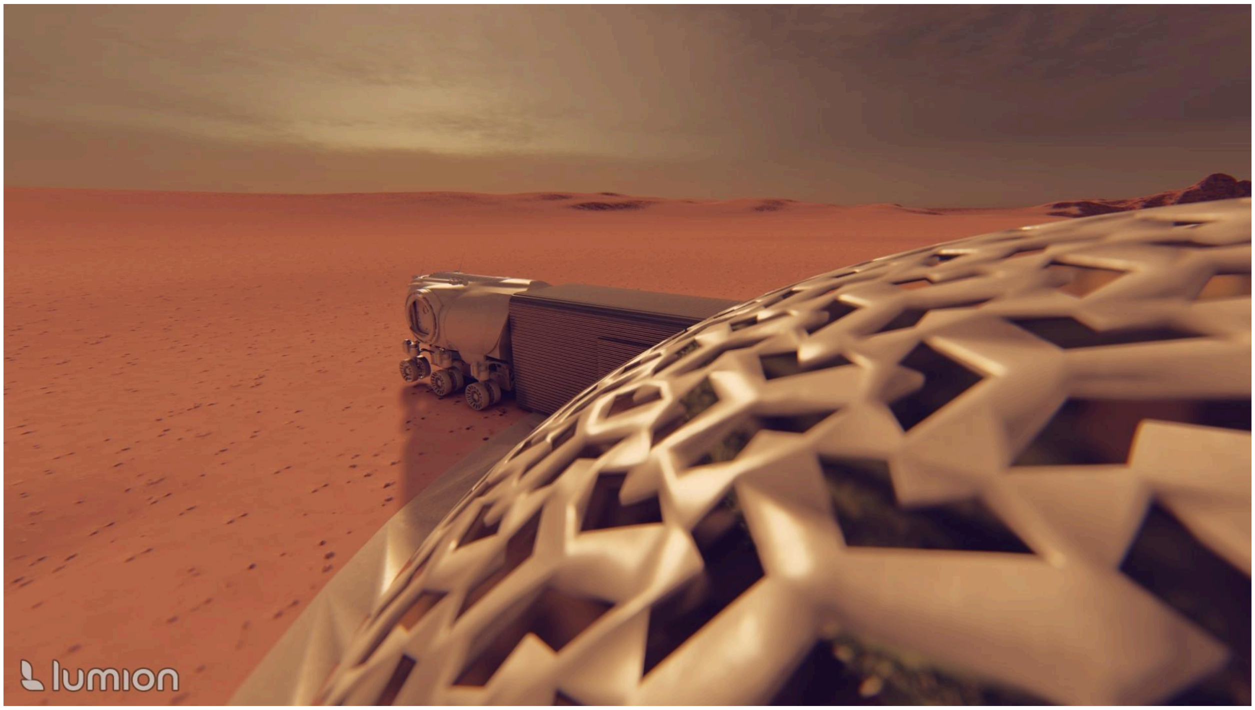
**Appendix H: Additional Image**







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